Lesson plan · for the teacher

One phone, many pupils

Programming · Level 4 · Lesson 2

Time
40 min

Materials
The phones you have, charged and prepared offline; A rota of roles on the board; Role cards: driver, navigator, tester, reporter

Learning goals

- Run pair programming and group roles, with swaps.
- Live-code from the front, with pupils predicting.
- Prepare phones to work offline.

Activity: Drivers and navigators

1. Live code (10 min): Write a short program on your phone or the board, line by line. Before each run, ask for predictions. Make one mistake on purpose and read the error together.
2. Roles (5 min): Hand out role cards: driver, navigator, tester, reporter. Write the rota on the board so everyone sees when they drive.
3. Debug (15 min): Each group gets the two-bug program on its phone (or on paper). Swap roles after each bug is fixed.
4. Report (5 min): Reporters explain one bug and how their group found it, using the error message.
5. Wrap-up (5 min): Check the rota: did everyone drive? Then give the exit ticket.

Prepare the phones the day before: lessons downloaded, a code step opened once, batteries charged.

Discussion

- How did the roles change who took part?
- What did pupils learn from your deliberate mistake?
- What will you prepare the day before every computing lesson?

Exit ticket

- Write your rota for the next lesson: groups, roles and swaps.

Try it at school tomorrow
Live-code a short program on your phone in front of the class (or write it on the board), pausing before each run: “What will it print?” Make one mistake on purpose and let pupils find it.

Practice questions · for you

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1 **Tomorrow's class has no internet. What must you do today?**

◯ Nothing: the app works anywhere ◯ Download the lessons and open a code step once

◯ Print every screen

2 **A program for a pair to fix: it should print the average of the marks typed, like Average: 12.0. It has two bugs. Driver types, navigator reads the errors; swap after the first fix.**

```
n = int(input("How many marks? "))
total = 0
for i in range(1, n):
    mark = input("Mark? ")
    total = total + mark
print("Average:", total / n)
```

Write your program:

3 **While live coding, you make this mistake on purpose. Which line should the class spot?**

Circle the line with the mistake. Write it correctly:

```
1  pupils = ["Amina", "Kofi", "Layla"]
2  for p in pupils
3      print("Good morning,", p)
```

4 **Live coding: you pause and ask the class to predict. What does this print?**

```
groups = 32 // 4
print("Groups of four:", groups)
```

◯ Groups of four: 8.0 ◯ Groups of four: 8 ◯ Groups of four: 32 // 4

5 **Match each role in a group of four to its job.**

Driver	• • • •	Tries inputs that might break it
Navigator	• • • •	Types or taps on the phone
Tester	• • • •	Explains the group's work to the class
Reporter	• • • •	Reads, checks and thinks ahead

How many devices can you count on for a lesson? Plan the roles: who drives, who navigates, and when do they swap? How will you make sure every pupil drives?

Check your answers

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1 **Download the lessons and open a code step once**

Downloaded lessons, plus one Python run while online, make everything work offline the next day.

2

```
n = int(input("How many marks? "))
total = 0
for i in range(n):
    mark = int(input("Mark? "))
    total = total + mark
print("Average:", total / n)
```

One bug gives an error (text added to a number); the other gives no error but a wrong answer (one mark too few).

3 **Line 2: `for p in pupils:`**

A for line needs a colon. Making a common mistake on purpose, then reading the error with the class, shows that errors are normal and how to read them.

4 **Groups of four: 8**

// divides and drops any remainder, giving a whole number: 8.

5 **Driver — Types or taps on the phone**
Navigator — Reads, checks and thinks ahead
Tester — Tries inputs that might break it
Reporter — Explains the group's work to the class

Four roles keep four pupils thinking about one program.